Round 1 5 categories · 100 to 500 pts each

Shape, Center & Spread

$100 When a distribution has a long tail stretching to the left and the mean is < the median, it has this shape.

Answer **What is left-skewed (negatively skewed)?**

Worked Solution

The tail stretches toward lower values, so the distribution is left-skewed. The mean is pulled in the direction of the tail, giving mean < median.

$200 For the data set 62, 68, 74, 80, 88, 95, 102, the interquartile range has this value.

Answer **What is 27?**

Worked Solution

Median = 80. Lower half: {62, 68, 74}, so Q1 = 68. Upper half: {88, 95, 102}, so Q3 = 95.

IQR = Q3 - Q1 = 95 - 68 = 27.

$300 Test scores are normally distributed with mean 72 and standard deviation 8. A student scoring 88 has this z-score.

Answer **What is z = 2?**

Worked Solution

z = (x - mean) / SD = (88 - 72) / 8 = 16 / 8 = 2.

The student scored 2 standard deviations above the mean.

$400 A boxplot shows Q1 = 20 and Q3 = 50. Using the 1.5 × IQR rule, a value of 100 would be identified as this.

Answer **What is an outlier?**

Worked Solution

IQR = 50 - 20 = 30. Upper fence = Q3 + 1.5(IQR) = 50 + 45 = 95.

Since 100 > 95, the value of 100 is an outlier.

$500 A data set of daily absences is: 0, 0, 1, 2, 2, 3, 24. The median is preferred over the mean as a measure of center for this reason.

Answer **What is the median is resistant to outliers (24 pulls the mean up but not the median)?**

Worked Solution

Mean = (0+0+1+2+2+3+24) / 7 = 32/7 ≈ 4.57. Median = 2.

The outlier 24 inflates the mean to 4.57, but 5 of 7 students had 2 or fewer absences. The median is resistant to outliers and better represents the typical value.

2-Var Quantitative Data

$100 A correlation coefficient of $r = -0.91$ indicates this about the relationship between two quantitative variables.

Answer **What is a strong, negative linear relationship?**

Worked Solution

Negative sign: as x increases, y tends to decrease. Magnitude 0.91 (close to 1) indicates a strong association.

$200 The LSRL for predicting exam score ($\hat{y}$) from hours studied (x) is $\hat{y} = 55 + 4.2x$. For each additional hour studied, the predicted exam score changes by this amount.

Answer **What is an increase of 4.2 points?**

Worked Solution

The slope (4.2) is the predicted change in y for each 1-unit increase in x. For each additional hour studied, the predicted exam score increases by 4.2 points.

$300 A regression model predicting college GPA from high school GPA has $r^2 = 0.46$. This percentage of the variation in college GPA is explained by its linear relationship with high school GPA.

Answer **What is 46%?**

Worked Solution

$r^2 = 0.46$: about 46% of the variation in college GPA is explained by the linear relationship with HS GPA. The remaining 54% is due to other factors.

$400 A student's actual exam score is 76, and the regression model predicted a score of 83. This student's residual equals this value, and the model did this.

Answer **What is -7; the model overpredicted?**

Worked Solution

Residual = Actual - Predicted = 76 - 83 = -7.

A negative residual means the model overpredicted (predicted higher than actual).

$500 A residual plot for a linear regression model shows a fan-shaped pattern, with the residuals spreading out as x increases. This indicates this problem with the model.

Answer **What is non-constant variance (the linear model may not be appropriate)?**

Worked Solution

Random scatter in a residual plot means the linear model is appropriate. A fan shape means the spread of residuals increases as x increases, indicating non-constant variance. The linear model may not be appropriate.

Bias & Sampling

$100 A researcher assigns every 10th person on an alphabetized list to participate in a survey. This sampling method is called this.

Answer **What is systematic random sampling?**

Worked Solution

Systematic sampling: select every kth person from a list (here, every 10th person) after a random starting point.

$200 A school reporter polls only students leaving the cafeteria about satisfaction with cafeteria food, then reports the findings as representative of the whole school. The main flaw is this type of bias.

Answer **What is undercoverage (or sampling bias)?**

Worked Solution

Students who do not eat in the cafeteria are excluded from the sample. Since only cafeteria-goers are surveyed, the sample does not represent all students. This is undercoverage.

$300 A polling company mails surveys to 5,000 households but only 800 respond. The results may be biased because responders may differ systematically from non-responders. This is called this type of bias.

Answer **What is non-response bias?**

Worked Solution

Non-response bias: people who choose to respond may differ systematically from those who do not. With only 800 of 5,000 responding, the results may not represent the full group.

$400 A researcher divides a city into zip-code areas, randomly selects 5 zip codes, and surveys every household in those zip codes. This sampling method is called this.

Answer **What is cluster sampling?**

Worked Solution

Cluster sampling: randomly select entire groups (zip codes here), then survey everyone in the selected groups. In stratified sampling, you randomly sample from within each group rather than taking all members.

$500 Researchers use the sample mean from 800 randomly selected carry-on bags to estimate mean weight. If they increase the sample size to 1,300 (all else equal), this describes what happens to the bias and variance of the estimator.

Answer **What is the bias stays the same and the variance decreases?**

Worked Solution

The sample mean is always unbiased: $E(\bar{x}) = \mu$ regardless of n. Bias stays at 0.

Variance = σ^2 / n. As n increases from 800 to 1,300, the variance decreases.

Experimental Design

$100 In a drug study, the group that receives an inactive substance designed to look like the real medication, to account for the placebo effect, is called this.

Answer **What is the control group (or placebo group)?**

Worked Solution

The control group receives a placebo and serves as the baseline. It accounts for the placebo effect: improvement that occurs simply because subjects believe they are being treated.

$200 The feature of a controlled experiment that allows researchers to draw cause-and-effect conclusions by balancing confounding variables between groups is this.

Answer **What is random assignment?**

Worked Solution

Random assignment distributes confounding variables roughly equally across groups by chance. Any difference in outcomes can then be attributed to the treatment, establishing causation.

$300 In a drug trial, neither the patients nor the evaluating physicians know who is receiving the drug or the placebo. This experimental design is called this.

Answer **What is double-blind?**

Worked Solution

Double-blind: both subjects and evaluators are unaware of who received the treatment. Subjects blinded prevents the placebo effect. Evaluators blinded prevents observer bias.

$400 A researcher plants 60 seeds: 30 randomly receive a new fertilizer and 30 receive none. After 8 weeks, mean growth is compared. The key feature that allows a causal conclusion about the fertilizer is this.

Answer **What is random assignment?**

Worked Solution

Random assignment ensures the two groups are comparable at the start, balancing initial differences by chance. If a significant growth difference is found, it can be attributed to the fertilizer.

$500 A researcher tests whether a sleep aid helps: each subject is tested once after the medication and once after a placebo, with the order randomized. Each subject acts as their own control in this experimental design.

Answer **What is a matched pairs design?**

Worked Solution

Matched pairs: each subject receives both treatments in random order and serves as their own control. The analysis examines the within-person difference (treatment result - placebo result), removing individual variability as a confounding factor.

Probability

$100 The probability of rolling a number > 4 on a single fair six-sided die is this.

Answer **What is 1/3 (approximately 0.333)?**

Worked Solution

Numbers > 4: {5, 6}. P(> 4) = 2/6 = 1/3 ≈ 0.333.

$200 A store's records show that 1/4 of customers who visit the service department ask for help finding an item. Assuming independence, the probability that at least 1 of the next 4 customers asks for help is this.

Answer **What is 1 - (3/4)^4 ≈ 0.684?**

Worked Solution

P(at least 1) = 1 - P(none ask for help) = 1 - (3/4)^4 = 1 - 81/256 ≈ 0.684.

\$300 In a class, 55% of students play a sport, 40% are in a school club, and 20% do both. The probability that a randomly selected student plays a sport or is in a club is this.

Answer **What is 0.75?**

Worked Solution

P(A or B) = P(A) + P(B) - P(A and B) = 0.55 + 0.40 - 0.20 = 0.75.

\$400 Events A and B have P(A) = 0.4, P(B) = 0.5, and P(A and B) = 0.20. This is the relationship between A and B.

Answer **What is independent (A and B are independent events)?**

Worked Solution

For independence, P(A and B) must equal P(A) × P(B). P(A) × P(B) = 0.4 × 0.5 = 0.20, which equals P(A and B) = 0.20. The events are independent.

\$500 According to a survey, 6% of workers arrive between 6:45 and 7:00 AM. If 300 workers are randomly selected and W is the number arriving in that window (independent arrivals), the standard deviation of W is closest to this value.

Answer **What is 4.11?**

Worked Solution

W ~ Binomial(n = 300, p = 0.06).

$SD = \sqrt{(np(1-p))} = \sqrt{(300 \times 0.06 \times 0.94)} = \sqrt{(16.92)} \approx 4.11.$